

FLADD: Federated Learning-based Privacy Protection for Audio Deepfake Detection

Authors: L. Zhang, X. Chen, and Y. Wang

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Abstract

Centralized audio deepfake detection model training raises privacy concerns when user speech recordings are uploaded to central cloud servers. FLADD introduces a privacy-preserving federated learning framework where client devices train local neural detector weights using differential privacy noise injection and communicate only model updates.

Key Methodology & Architecture

Federated Averaging (FedAvg), Differential Privacy (DP-SGD), Light Convolutional Neural Network (LCNN), secure aggregation protocol.

Datasets & Benchmark Environments

ASVspoof 2019, LibriSpeech, Local Edge Device Testbeds.

Performance Metrics & Graph Axis Parameters

Reported Results: Global Model Accuracy = 96.85%, Differential Privacy Epsilon (ϵ) = 2.1, EER = 1.45%.

Graph Parameter Mapping: X-axis: Federated Training Rounds (0 to 100 rounds); Y-axis: Global Model Test Accuracy (% from 50% to 100%).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Guarantees user voice privacy while building robust collaborative detection models.	Communication overhead over multi-client mobile networks.

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